

Cooperativity-Induced Bifurcation and Phase Separation in Stress-Driven RNA–Peptide Systems

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Environmental stress can act as a selective pressure that drives cooperative coupling in RNA–peptide mixtures. We develop a minimal coevolutionary model in which stress simultaneously accelerates degradation and favors protective heterotypic interactions. Through controlled adiabatic elimination of fast population and trait variables, we obtain a one-dimensional dynamical equation for a dimensionless coupling coordinate X . The model reveals a cooperativity-induced saddle-node bifurcation that occurs only when the protection kernel has Hill exponent $n > 1$. Below a critical stress η_c , the system resides in a low-coupling state; above η_c , it switches discontinuously to a high-coupling branch with asymptotic scaling $X^* \sim (nA\eta)^{1/(n+1)}$. This bifurcation is accompanied by complete decoupling between functional evolution (slow) and total molecular abundance (fast), a generic consequence of timescale separation under resource limitation. We further relate X to an effective Flory–Huggins interaction parameter through a bounded monotone mapping, showing that the dynamical jump can bring the mixture close to a mean-field spinodal for liquid–liquid phase separation. Stochastic Monte Carlo simulations of a particle-based RNA–peptide system confirm that the reduced description captures the dominant coarse-grained dynamics. Our results demonstrate how cooperative feedback can convert a continuously increasing stress into a qualitative transition in both molecular function and mesoscale organization.

I. INTRODUCTION

Environmental stress is often regarded as a purely destructive influence. In evolving molecular systems, however, stress can also act as a selective filter that favors more strongly coupled and more resilient molecular states. This idea is particularly natural in RNA–peptide mixtures, where peptides can protect RNA from degradation, stabilize catalytic function, and enhance effective replication, while RNA can in turn support peptide retention or production. In such systems, function and organization are not independent: the same heterotypic interactions that improve survival may also push the mixture toward condensation.

A central question is therefore whether stress merely shifts molecular performance continuously, or whether it can induce a qualitative transition in the dynamical state of the system. If protection is cooperative, then increasing stress need not move the system smoothly along a single response branch. Instead, it may trigger a bifurcation between low- and high-coupling states. In the present framework this distinction is controlled by the Hill exponent n , which is interpreted as an effective cooperativity index rather than a decorative fitting parameter.

The second question concerns organization. If adaptive coupling strengthens productive RNA–peptide contacts, then the same evolutionary variable that controls function may also be used, at the coarse-grained level, to parameterize an effective heterotypic interaction strength in a thermodynamic description. This motivates a two-step scenario: stress first drives the system toward a high-coupling dynamical state, and the resulting change in the coarse-grained interaction parameter may bring the mixture closer to a spinodal threshold associated with liquid–liquid phase separation (LLPS). In this work, however,

that thermodynamic link is used as a phenomenological consistency check rather than as a microscopic derivation of phase separation.

In this work we construct and analyze such a minimal model. We first formulate stress-dependent replication and degradation kinetics at the mean-field level, then perform a controlled quasi-steady reduction to a single dimensionless coupling variable X . We show that cooperative protection ($n > 1$) induces a saddle-node bifurcation, leading to a discontinuous jump in coupling strength under increasing stress. The same variable X can be mapped phenomenologically to an effective heterotypic interaction parameter in a Flory–Huggins description, suggesting a plausible route by which stress-driven adaptation may facilitate phase separation.

The reduction distinguishes two logically distinct steps: (i) an asymptotically justified elimination of fast variables based on explicit small parameters, and (ii) a minimal closure for the remaining slow dynamics. Quantitative details such as the precise location of the bifurcation depend on the chosen closure, but the existence of a cooperativity-driven qualitative transition is a robust structural feature of the reduced kinetics. This framework provides clear, testable predictions for in vitro RNA–peptide systems and highlights the interplay between nonequilibrium adaptation and thermodynamic instability in prebiotic chemistry.

II. MODEL

We consider a well-mixed RNA–peptide mixture described by the total RNA concentration $R(t)$ and peptide concentration $P(t)$. At a more microscopic level, RNAs and peptides are assumed to carry continuous

traits x and y characterizing, respectively, RNA binding propensity and peptide protective or catalytic effectiveness. Their mean values are denoted by $\bar{x}(t)$ and $\bar{y}(t)$. To eliminate the dimensional ambiguity present in the earlier draft, we take both traits to be dimensionless quantities:

$$[x] = [y] = 1, \quad (1)$$

while concentrations have dimension C and time has dimension T .

The replication rate is written as

$$r(\bar{x}, t) = \alpha[1 + \beta \bar{x} P(t) \bar{y}(t)], \quad (2)$$

where α has dimension T^{-1} and β has dimension C^{-1} . Environmental stress enhances degradation but this damage is mitigated by cooperative RNA-peptide protection. We therefore introduce the protection occupancy

$$\Theta(c(t)) = \frac{c(t)^n}{1 + c(t)^n}, \quad c(t) = \gamma \bar{x}(t) P(t) \bar{y}(t), \quad (3)$$

where γ has dimension C^{-1} , so that $c(t)$ is dimensionless. The stress-dependent degradation rate becomes

$$d(\bar{x}, t) = \delta_0 E [1 - \Theta(c)] = \frac{\delta_0 E}{1 + c(t)^n}, \quad (4)$$

with δ_0 of dimension T^{-1} and E dimensionless.

Equation (4) may also be viewed as the coarse-grained limit of a multivalent protection process. If efficient shielding requires the formation of n cooperative RNA-peptide contacts within a local complex, then one may introduce a local association variable c_{loc} together with an occupancy

$$\Theta(c_{\text{loc}}) = \frac{c_{\text{loc}}^n}{1 + c_{\text{loc}}^n}.$$

In the present nondimensionalized formulation, the corresponding dissociation scale has been absorbed into γ , so that c_{loc} is represented directly by the dimensionless combination $c(t) = \gamma \bar{x}(t) P(t) \bar{y}(t)$.

For the family considered in this manuscript we restrict to $n \geq 1$. The case $n = 1$ defines the noncooperative baseline: the protection response is smooth and monotone, and the resulting reduced dynamics are monostable. By contrast, the cooperative regime $n > 1$ produces an interior maximum in the corresponding feedback kernel and is therefore the only regime in which a saddle-node bifurcation can occur. Values $n < 1$ are not considered here, since they generate a singular response at small X in the present Hill-type closure and lie outside the intended coarse-grained description.

To incorporate finite resources, we supplement the molecular rates by logistic regulation:

$$\frac{dR}{dt} = [r(\bar{x}, t) - d(\bar{x}, t)] R \left(1 - \frac{R}{K_R}\right), \quad (5)$$

$$\frac{dP}{dt} = [k_0 R + \kappa R \bar{x} P \bar{y}] \left(1 - \frac{P}{K_P}\right) - \mu P, \quad (6)$$

where K_R and K_P are carrying capacities with dimension C , k_0 and μ have dimension T^{-1} , and κ has dimension $C^{-1}T^{-1}$. The nonlinear term $\kappa R \bar{x} P \bar{y}$ describes autocatalytic peptide production: existing high-quality peptides (trait $\bar{y}$) act as cofactors to promote ribozyme-catalyzed peptide synthesis, mediated by RNA binding affinity $\bar{x}$, consistent with prebiotic scenarios of coupled RNA-peptide coevolution. Equation (5) assumes positive net growth $r(\bar{x}, t) > d(\bar{x}, t)$; if net growth is negative, R decays to 0 and adaptation cannot proceed, so our analysis is restricted to regimes where molecular survival is possible.

The adiabatic slaving relation $\bar{y} \simeq \lambda \bar{x}$ follows as the leading-order quasi-steady solution of the peptide mean-trait closure

$$\frac{d\bar{y}}{dt} = \kappa V_y R \bar{x} - \mu_y \bar{y}, \quad (7)$$

with the trait origin shifted so that the baseline state is $x_0 = y_0 = 0$. Here V_y denotes the dimensionless variance of the peptide-trait distribution. In the singular-perturbation regime

$$\varepsilon_P \equiv \frac{\mu_x}{\alpha} \ll 1, \quad \varepsilon_y \equiv \frac{\mu_x}{\mu_y} \ll 1, \quad (8)$$

the concentrations relax rapidly to their saturation values, $R = K_R[1 + O(\varepsilon_P)]$ and $P = K_P[1 + O(\varepsilon_P)]$. The quasi-steady approximation then gives

$$\bar{y} = \frac{\kappa V_y}{\mu_y} R \bar{x} + O(\varepsilon_y) = \lambda \bar{x} + O(\varepsilon_y + \varepsilon_P), \quad \lambda \equiv \frac{\kappa V_y K_R}{\mu_y}. \quad (9)$$

Thus the reduction of the full kinetic system is controlled by explicit small parameters, while the final scalar equation introduced below is a minimal coarse-grained closure that retains the dominant cooperative feedback at the level of the slow observable X . The detailed moment-closure derivation is given in Appendix C, and the effective scalar reduction is recovered in Appendix D. This reduction holds in a well-defined singular-perturbation regime where population and trait turnover timescales are hierarchically separated; the formal dimensionless small parameters and validity criteria are derived explicitly in Sec. III. We later introduce a dimensionless reduced coordinate X (formally defined in Sec. III) that absorbs this bilinear $x \times y$ interaction; any smooth monotone reparametrization of $\bar{x}$ would generate the same phase portrait, and X is chosen for analytical convenience to yield a closed scalar equation with strictly dimensionless control parameters. To further clarify the separation of dynamical and thermodynamic roles, we define a *dynamical order parameter* for functional coupling:

$$\Phi(t) \equiv R(t) P(t) \bar{x}(t) \bar{y}(t), \quad (10)$$

which quantifies the total effective heterotypic contact capacity of the system. Under the resource-saturated quasi-steady assumption $R \simeq K_R$ and $P \simeq K_P$, this

reduces to

$$\Phi \simeq K_R K_P \lambda \bar{x}^2 = \frac{K_R}{\gamma} X, \quad (11)$$

explicitly linking the reduced evolutionary variable X to the macroscopic interaction capacity. Critically, Φ is a dynamical quantity shaped by evolution; the thermodynamic phase behavior is governed by a separate effective interaction parameter that depends monotonically on Φ , as derived in later sections.

III. DIMENSIONLESS REDUCTION

We now formalize the controlled asymptotic reduction of the full kinetic model, starting from the timescale hierarchy that justifies adiabatic elimination of fast degrees of freedom. In the limit of well-separated timescales, the fast population variables relax to their saturated values and the peptide trait follows its quasi-steady manifold:

$$R = K_R + O(\varepsilon_P), \quad P = K_P + O(\varepsilon_P), \quad \bar{y} = \lambda \bar{x} + O(\varepsilon_P + \varepsilon_d), \quad (12)$$

Substituting these leading-order relations into the moment-closure equations yields the one-dimensional reduced dynamics for X .

This quasi-steady reduction is valid only when three dimensionless parameters satisfy smallness criteria, corresponding to the three core approximations of the framework:

$$\varepsilon_P \equiv \frac{\mu_x}{\alpha} \ll 1, \quad (13)$$

$$\varepsilon_y \equiv \frac{\mu_x}{\mu_y} \ll 1, \quad (14)$$

$$\varepsilon_V \equiv \frac{V_x}{\langle x^2 \rangle} \ll 1, \quad (15)$$

Here ε_P measures the ratio of the RNA-trait evolutionary rate to the population growth rate; $\varepsilon_P \ll 1$ ensures that the total RNA and peptide concentrations relax to their carrying capacities on a timescale much shorter than that of trait evolution, justifying $R \simeq K_R$ and $P \simeq K_P$. Similarly, $\varepsilon_y \ll 1$ guarantees that the peptide mean trait $\bar{y}$ equilibrates to its adiabatic value $\lambda \bar{x}$ before the RNA mean trait $\bar{x}$ changes appreciably. In typical prebiotic scenarios, the replication rate α is of order 1 h^{-1} , the peptide turnover rate μ_y is of order 10^{-1} h^{-1} , and the RNA trait relaxation rate μ_x is of order 10^{-4} h^{-1} ; hence $\varepsilon_P \sim 10^{-4}$ and $\varepsilon_y \sim 10^{-3}$, both satisfying the required inequalities.

The third criterion $\varepsilon_V \ll 1$ enforces the *narrow-trait-distribution approximation* used to close the moment hierarchy in Appendix B: the relative variance of the RNA trait distribution must be small enough that higher-order moments do not contribute appreciably to the mean adaptive dynamics. For a distribution with mean $\bar{x}$ and variance V_x , $\langle x^2 \rangle = \bar{x}^2 + V_x$, so $\varepsilon_V \approx V_x / \bar{x}^2$ for $\bar{x} \gg \sqrt{V_x}$. In the baseline parameterization, we take $V_x = 0.01$ and

$\bar{x} \sim 0.1 - 1$, giving $\varepsilon_V \sim 0.01 - 1$, which holds far from bifurcation thresholds. However, as shown in the following section, this condition can break down near the saddle-node bifurcation point, where strong selection gradients broaden the trait distribution.

If any of the three conditions fails (e.g., under extremely slow growth, very fast RNA trait mutation, or near a critical bifurcation point), the one-dimensional reduction ceases to be quantitatively accurate, and the full moment-closure system (including evolving trait variances) or a trait-resolved model must be retained.

The composite coupling entering the protection kernel in Eq. (4) is

$$c(t) \simeq \gamma K_P \lambda \bar{x}^2. \quad (16)$$

This motivates the dimensionless state variable

$$X \equiv \gamma K_P \lambda \bar{x}^2, \quad (17)$$

which is a coarse-grained coupling coordinate rather than a literal microscopic bond number. In the present reduction, X is the slow variable retained after adiabatic elimination of fast degrees of freedom; experimentally, it may be proxied by normalized stress survival, heterotypic retention efficiency, or contact-enrichment observables. The interpretation of X is therefore phenomenological but dimensionally consistent.

We then introduce the reduced time

$$\tau \equiv \mu_x t, \quad (18)$$

and define the two dimensionless control parameters

$$\mathcal{A} \equiv \frac{\alpha \beta V_x K_P \lambda}{\mu_x}, \quad \eta \equiv \frac{\gamma \delta_0 E}{\alpha \beta}, \quad (19)$$

where V_x is the dimensionless variance of the RNA trait distribution. Here $\mathcal{A}$ measures the baseline adaptive feedback strength and η measures the dimensionless stress load. Their dimensionlessness follows directly from the assigned dimensions listed in Table II; no additional rescaling is introduced at this stage.

With these definitions, the reduced one-dimensional dynamics are written in the following minimal closed form:

$$\frac{dX}{d\tau} = 2X \left[\mathcal{A} \left(1 + \eta \frac{nX^{n-1}}{(1+X^n)^2} \right) - 1 \right]. \quad (20)$$

Equation (20) is the effective scalar closure of the asymptotically reduced model. The parameters $\mathcal{A}$ and η are strict control parameters, while n quantifies cooperativity in the adopted Hill-type closure. The reduction fixes the slow variable and the parameter scaling, but the specific functional form of the feedback kernel is a modeling choice chosen to provide the simplest bounded cooperative response consistent with saturation.

The preceding derivation therefore distinguishes two steps: (i) a controlled elimination of fast variables, which

is asymptotically justified, and (ii) the closure of the residual scalar dynamics, which is phenomenological at the coarse-grained level. The particular choice

$$\frac{nX^{n-1}}{(1+X^n)^2}$$

is introduced as the simplest analytic kernel that produces cooperative switching while remaining bounded for large X . Its role is to retain the dominant collective trend of the coarse-grained observable, not to encode a unique microscopic binding law. Accordingly, the bifurcation analysis below is exact for the closed equation written here, while its microscopic interpretation is limited to the closure assumptions spelled out above.

A. Population saturation and the survival threshold

The quasi-steady approximation $R \simeq K_R$ and $P \simeq K_P$ assumes not only a separation of timescales ($\varepsilon_P \ll 1$) but also a positive net growth rate $r - d > 0$ that allows the populations to reach their carrying capacities. Without protection ($\bar{x} = \bar{y} = 0$), the net growth rate reduces to

$$r - d = \alpha - \delta_0 E. \quad (21)$$

A necessary (though not sufficient) condition for the population to avoid immediate collapse is therefore $\alpha > \delta_0 E$. Using the definition $\eta = \gamma \delta_0 E / (\alpha \beta)$, this inequality translates into a strict bound on the dimensionless stress parameter:

$$\eta < \frac{\gamma}{\beta}. \quad (22)$$

If η exceeds γ/β , the unprotected system suffers a negative net growth rate, the total RNA concentration decays below K_R , and the resource-saturated reduction $R \simeq K_R$ cannot be reached from a low-coupling initial condition. Even if the adaptive feedback could eventually restore a positive net growth through a sufficiently large X , the transient collapse would invalidate the timescale hierarchy and require a fully coupled population–trait description. Throughout this paper we therefore restrict the analysis to the regime $\eta < \gamma/\beta$, where the saturated quasi-steady state is accessible for all coupling values.

For finite coupling $X > 0$, the net growth rate is given by

$$r - d = \alpha \left[1 + \frac{\beta}{\gamma} X \right] - \frac{\delta_0 E}{1 + X^n}, \quad (23)$$

where we have used $X = \gamma K_P \lambda \bar{x}^2$ and the relation $\beta \bar{x} P \bar{y} = (\beta/\gamma)X$ that follows from the saturated quasi-steady conditions. At the onset of the high-coupling branch ($X \sim 1$ for $\eta \approx \eta_c$), the protection term reduces the effective degradation to $\sim \delta_0 E/2$, while the replication term is enhanced by a factor $\sim 1 + \beta/\gamma$. Hence, if the

bare condition $\eta < \gamma/\beta$ is satisfied, the net growth rate remains comfortably positive during the whole adaptive trajectory. Should η approach γ/β , however, ε_P ceases to be small because the population relaxation time $1/(r-d)$ becomes comparable to the trait evolution time $1/\mu_x$, and the one-dimensional reduction must be replaced by the full population–trait system.

The bound (22) provides a concrete, parameter-free estimate of the stress range over which the adiabatic elimination of the total concentrations is self-consistent. In the numerical illustrations we keep η well below this threshold, ensuring that the conclusions drawn from the reduced model are not artifacts of an unjustified saturation ansatz.

B. Validity of Moment Closure Near Bifurcation Thresholds

The narrow-trait-distribution assumption underpinning the scalar reduction assumes that the selection gradient $\partial(r-d)/\partial x$ is sufficiently weak that it does not significantly broaden the trait distribution. Near the saddle-node bifurcation threshold, however, the nonlinearity of the cooperative protection kernel creates a sharp selection pressure favoring higher-coupling traits, which can broaden the trait distribution, increase ε_V , and introduce quantitative deviations from the reduced model predictions.

To quantify this effect, we compare the steady-state coupling X^* predicted by the one-dimensional reduced model (which assumes fixed, small V_x) to that obtained from a *full second-order moment-closure model*, which evolves both the mean trait $\bar{x}$ and the trait variance V_x self-consistently (see Appendix B1 for the full system of equations). The relative deviation between the two predictions is defined as

$$\Delta_X^* = \frac{|X_{\text{full}}^* - X_{\text{reduced}}^*|}{1 + X_{\text{reduced}}^*}, \quad (24)$$

consistent with the error metric used for stochastic simulations in Sec. ??.

Figure ?? shows the dependence of Δ_X^* on the dimensionless stress η for a representative parameter set with $n = 2$, $\mathcal{A} = 0.5$, and three values of the baseline mutational variance D_x (which sets the minimum trait variance in the absence of selection). For all parameter sets, the deviation is small ($\Delta_X^* < 5\%$) far from the critical stress η_c , but rises to a sharp peak near $\eta = \eta_c$, where the selection gradient is maximized. For baseline variances consistent with low mutation rates ($D_x = 10^{-4}$), the deviation remains below 10% for all η outside a narrow window of width $\Delta\eta \approx 0.1\eta_c$ around the bifurcation point. For higher mutation rates (broader baseline distributions), the deviation increases, and the region of validity of the scalar reduction shrinks further.

These results imply that the one-dimensional reduced model remains quantitatively accurate for most biolog-

ically relevant parameter regimes, except very close to the saddle-node threshold where higher-moment effects become non-negligible. The qualitative bifurcation structure (presence of bistability, scaling of the high-coupling branch) remains robust even in this region, as confirmed by trait-resolved simulations (Appendix E), but quantitative predictions of critical thresholds should include higher-moment corrections if precision near η_c is required.

C. Hierarchy of derivation, closure assumptions, and robustness

To make the logical structure of the reduction explicit, it is useful to distinguish three levels of approximation.

a. (i) Structure inherited from the reduced kinetic model. The following elements follow from the original dynamical model after timescale separation, nondimensionalization, and elimination of the fast variables:

1. the existence of a slow effective variable X that parameterizes the residual adaptive dynamics;
2. the appearance of a dimensionless feedback parameter $\mathcal{A}$;
3. the appearance of a dimensionless stress parameter η ;
4. the generic multiplicative structure of the reduced equation, in which the growth rate of X is proportional to X itself and is modulated by a nonlinear feedback factor.

These features reflect the reduced kinetic architecture and do not depend on the specific Hill form used below.

b. (ii) Closure choices at the coarse-grained level. To obtain a closed scalar equation that is analytically tractable, we adopt the following minimal closure choices:

1. *Hill-type cooperative kernel.* The protection feedback is represented by

$$h_n(X) = \frac{nX^{n-1}}{(1 + X^n)^2}, \quad (25)$$

which is the simplest bounded saturating form that captures cooperative response and an interior maximum for $n > 1$.

2. *Monotone saturating thermodynamic mapping.* The effective interaction parameter is written as

$$\chi_{\text{eff}}(X) = \chi_0 + \Delta\chi\psi(X), \quad (26)$$

where $\psi(X)$ is assumed to be monotone and bounded.

3. *Lowest-order moment closure.* The RNA and peptide trait distributions are assumed to be narrow ($\varepsilon_V \ll 1$), so higher moments (including evolving

trait variances) are neglected beyond the leading constant-variance closure level used to define X . This assumption breaks down near the saddle-node bifurcation threshold, as discussed in Sec. III B.

These assumptions are not uniquely imposed by the microscopic model; rather, they provide the simplest analytically controlled closure consistent with cooperative feedback and saturation.

c. (iii) Results robust to broad smooth kernels. Several qualitative conclusions follow from the reduced structure and remain valid for a broad class of smooth cooperative-saturating kernels:

1. a threshold separating extinction-like and coupled states;
2. the possibility of bistability when the feedback kernel develops an interior maximum;
3. the existence of a saddle-node bifurcation as the generic local mechanism for switching between low- and high-coupling branches;
4. the interpretation of X as a coarse-grained adaptive coordinate.

These conclusions do not rely on the exact Hill numerator and denominator separately, but only on the existence of a nonlinear feedback with a single interior extremum.

d. (iv) Closure-dependent quantitative predictions. By contrast, the following predictions depend on the chosen closure:

1. the exact location of the saddle-node threshold;
2. the explicit expression for the critical stress η_c ;
3. the asymptotic scaling law on the high-coupling branch;
4. the quantitative mapping between X and χ_{eff} ;
5. the precise location of the spinodal boundary inferred from $\chi_{\text{eff}}(X)$.

Accordingly, these quantities should be interpreted as predictions of the chosen minimal closure, not as universal consequences of the microscopic model.

e. (v) Interpretation. In summary, the present reduction separates a robust dynamical backbone from model-dependent quantitative closure. The existence of cooperative switching is a structural result of the reduced kinetics, whereas the numerical bifurcation thresholds and the mapping to LLPS depend on the particular Hill kernel and saturating thermodynamic closure adopted here.

IV. COOPERATIVITY-INDUCED BIFURCATION

The steady states of Eq. (20) satisfy either $X = 0$ or

$$\mathcal{G}(X) = 1, \quad \mathcal{G}(X) \equiv \mathcal{A}[1 + \eta h_n(X)], \quad (27)$$

with

$$h_n(X) = \frac{nX^{n-1}}{(1+X^n)^2}. \quad (28)$$

For the present closure we restrict to $n \geq 1$. The noncooperative baseline corresponds to $n = 1$, for which

$$h_1(X) = \frac{1}{(1+X)^2},$$

so $h_1(X)$ is strictly decreasing for $X > 0$. Consequently, $\mathcal{G}(X)$ is also strictly decreasing, and the reduced dynamics are monostable: the $n = 1$ case does not admit a saddle-node bifurcation. The saddle-node analysis below therefore applies only to the cooperative regime $n > 1$, where $h_n(X)$ becomes unimodal and develops an interior maximum.

A saddle-node bifurcation requires the simultaneous conditions

$$\mathcal{G}(X) = 1, \quad \frac{d\mathcal{G}}{dX} = 0. \quad (29)$$

Since $\mathcal{G}'(X) \propto h'_n(X)$, the extremum of the nonlinear feedback term is determined by

$$X_{\text{sn}}^n = \frac{n-1}{n+1}, \quad n > 1. \quad (30)$$

Evaluating Eq. (28) at this point gives

$$C_n = \frac{(n+1)^2}{4n} \left(\frac{n-1}{n+1} \right)^{\frac{n-1}{n}}. \quad (31)$$

The criticality condition therefore becomes

$$\mathcal{A}(1 + \eta C_n) = 1, \quad (32)$$

which yields the critical stress

$$\eta_c = \frac{1/\mathcal{A} - 1}{C_n}, \quad n > 1. \quad (33)$$

This threshold is physically meaningful only for $0 < \mathcal{A} < 1$. For $\mathcal{A} \geq 1$, the high-coupling state is already accessible at zero stress, so no stress-induced bifurcation is required. For $0 < \mathcal{A} < 1$, increasing η first creates a pair of nonzero fixed points at the saddle-node; the intermediate branch is unstable, while the outer branches are stable by the standard one-dimensional phase-line argument.

a. Scope of the bifurcation criterion. The saddle-node condition above is a property of the specific Hill-type closure used in Eq. (28) and of the quasi-steady one-dimensional reduction that leads to Eq. (20). It should not be applied to the noncooperative baseline $n = 1$, for which the reduced dynamics are monostable. More generally, the same local bifurcation mechanism would appear for any smooth bounded kernel $q(X)$ with a single interior maximum q_{max} , in which case the onset criterion takes the generic form

$$\mathcal{A}(1 + \eta q_{\text{max}}) = 1, \quad (34)$$

but the numerical value of η_c depends on the chosen closure.

b. Interpretation. The bifurcation is controlled by cooperativity in the protection kernel, not by an arbitrary nonlinear fit. The Hill exponent n therefore plays a genuine mechanistic role: it distinguishes the smooth monostable response at $n = 1$ from the switch-like regime $n > 1$ in which a saddle-node bifurcation becomes possible. In short, the noncooperative limit is a separate baseline theory, whereas the saddle-node analysis is reserved for the cooperative regime.

V. FUNCTIONAL-ABUNDANCE DECOUPLING UNDER RESOURCE LIMITATION

The timescale hierarchy that underpins the one-dimensional reduction (i.e., $\varepsilon_P \ll 1$, $\varepsilon_y \ll 1$ from Sec. III [Sec. III]) directly leads to a biologically critical phenomenon: functional evolution is fully decoupled from population abundance under resource limitation. This decoupling operates on two distinct levels:

a. Temporal decoupling. Population abundance dynamics occur on the fast timescale set by replication and degradation ($\tau_{\text{growth}} \sim 1/\alpha$), while trait evolution proceeds on the slow timescale set by mutation and selection ($\tau_{\text{evo}} \sim 1/\mu_x \gg \tau_{\text{growth}}$). As a result, total RNA and peptide concentrations rapidly saturate to their carrying capacities $R \simeq K_R$, $P \simeq K_P$ within a few replication cycles, while the effective coupling strength X (and associated functional traits $\bar{x}, \bar{y}$) continues to evolve over orders of magnitude longer timescales without any further change in total molecular abundance. This temporal separation is directly testable in vitro: measuring total RNA/peptide concentrations and functional metrics (e.g., RNA stress survival rate, peptide catalytic efficiency) over time will show abundances plateau first, followed by sustained functional improvement.

b. Parameter decoupling. The steady-state coupling strength X^* depends only on the fixed carrying capacities K_R, K_P embedded in the dimensionless feedback parameter $\mathcal{A}$, not on transient fluctuations of R or P . Recalling the definition of $\mathcal{A}$ from Eq. (19):

$$\mathcal{A} = \frac{\alpha\beta V_x K_P \lambda}{\mu_x}, \quad \lambda = \frac{\kappa V_y K_R}{\mu_y} \implies \mathcal{A} \propto K_R K_P, \quad (35)$$

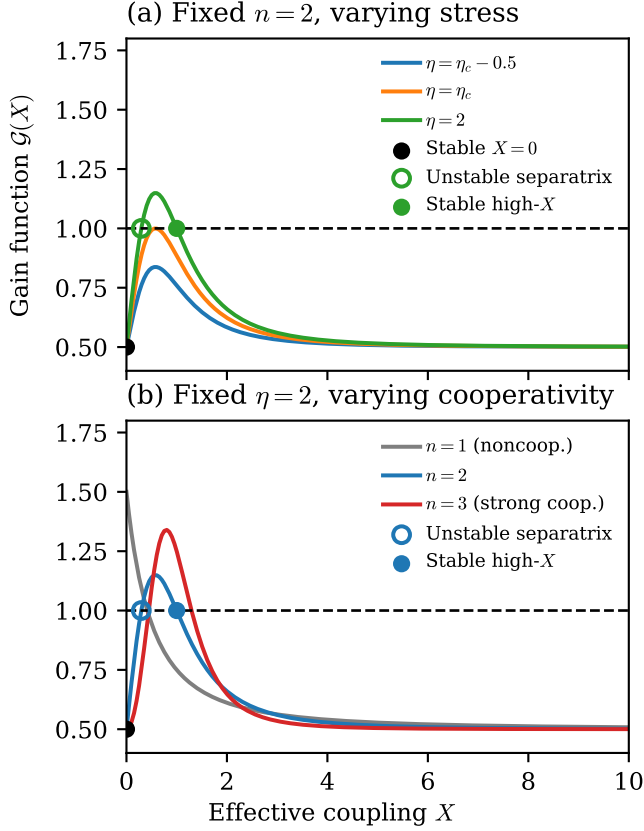


FIG. 1. Gain function $\mathcal{G}(X)$ [Eq. (27)] that determines the steady states of the reduced dynamics [Eq. (20)]. The horizontal dotted line indicates the fixed-point condition $\mathcal{G}(X) = 1$. (a) Fixed cooperativity $n = 2$ and baseline adaptive feedback $\mathcal{A} = 0.5$ for three values of dimensionless stress η : below the saddle-node threshold ($\eta = \eta_c - 0.5$, blue), exactly at the bifurcation ($\eta = \eta_c$, orange), and above threshold ($\eta = 2$, green). Filled circles mark stable fixed points ($X = 0$ and the high-coupling branch); the open circle denotes the unstable separatrix. (b) Fixed stress $\eta = 2$ and $\mathcal{A} = 0.5$ for three values of the cooperativity exponent n : $n = 1$ (noncooperative, gray), $n = 2$ (blue), and $n = 3$ (strong cooperation, red). For $n > 1$ the gain function develops an interior maximum, producing bistability with an unstable low-coupling separatrix (open blue circle) and a stable high-coupling branch (filled blue circle).

the steady-state coupling scales with resource availability as:

$$X^* \sim (n\mathcal{A}\eta)^{1/(n+1)} \sim (K_R K_P E)^{1/(n+1)} \quad (36)$$

in the high-stress limit (from Eq. (37)). This means richer prebiotic environments (higher nucleotide/amino acid availability, larger K_R, K_P) support higher ultimate functional coupling for a given stress level, but functional improvement does not require continuous growth of total population size.

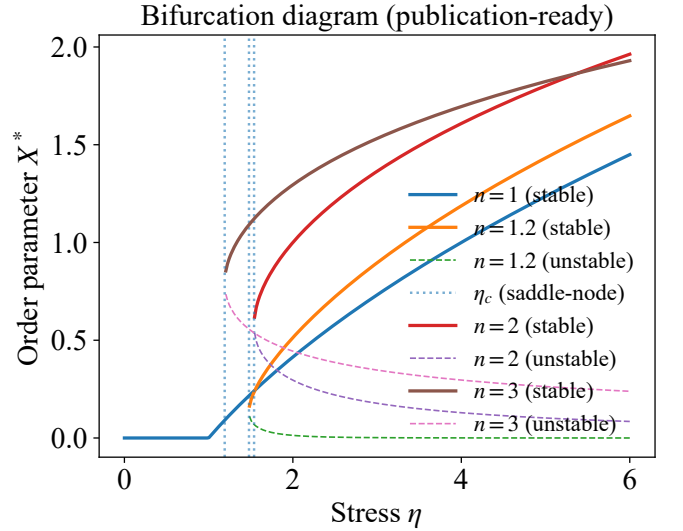


FIG. 2. Bifurcation structure of the reduced model as a function of the dimensionless stress η . Each point represents a steady-state solution X^* obtained numerically. For $n = 1$, the system remains monostable for all η . For $n > 1$, multiple steady states emerge, indicating the onset of bistability via a saddle-node bifurcation. This demonstrates that cooperativity ($n > 1$) is required for the emergence of nonlinear switch-like behavior in the stress-response dynamics.

VI. ASYMPTOTIC SCALING OF THE HIGH-COUPLING BRANCH

In the strong-stress regime, the nontrivial steady state can be estimated analytically. If $\eta \gg 1$ and the positive branch lies at $X \gg 1$, Eq. (27) implies

$$\mathcal{A}\eta \frac{n}{X^{n+1}} \sim 1 - \mathcal{A}, \quad \text{so that} \quad X_* \sim \left(\frac{n\mathcal{A}\eta}{1 - \mathcal{A}} \right)^{1/(n+1)}. \quad (37)$$

The cooperativity exponent therefore affects not only the existence of a bifurcation but also the scaling law of the high-coupling branch itself.

VII. MONOSTABLE GLOBAL STABILITY

In the monostable regime, where $\mathcal{G}(X)$ is strictly decreasing for $X > 0$ and the positive fixed point $X_* > 0$ is unique, global asymptotic stability follows from a Lyapunov argument. Consider

$$\mathcal{V}(X) = \frac{1}{2}(X - X_*)^2. \quad (38)$$

Using Eq. (20) yields

$$\frac{d\mathcal{V}}{d\tau} = 2X(X - X_*)[\mathcal{G}(X) - 1]. \quad (39)$$

Because $\mathcal{G}(X)$ is strictly decreasing for $X > 0$ in the monostable regime, $\mathcal{G}(X) - 1$ is positive when $X < X_*$

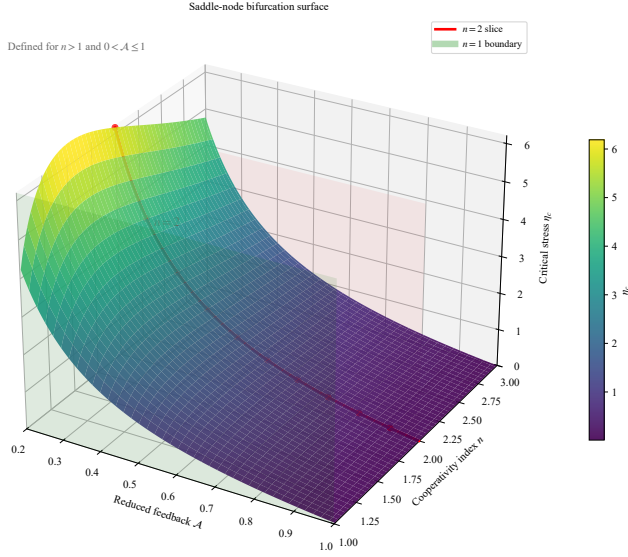


FIG. 3. Saddle-node bifurcation surface in the parameter space $(\mathcal{A}, n, \eta)$ defined by Eq. (32). The surface exists only for $n > 1$ and $0 < \mathcal{A} \leq 1$, where the nonlinear feedback develops an internal maximum. The red curve shows a representative slice at $n = 2$. For fixed cooperativity $n > 1$, decreasing $\mathcal{A}$ increases the critical stress η_c , while stronger cooperativity (larger n) lowers the threshold for the onset of bistability. The translucent green plane indicates the boundary $n = 1$, below which no saddle-node bifurcation occurs.

and negative when $X > X_*$, so it has the same sign as $X_* - X$. Substituting into Eq. (39) yields

$$\frac{dV}{d\tau} \leq 0, \quad (40)$$

with equality only at $X = X_*$. Hence X_* is globally asymptotically stable for all positive initial conditions in the monostable region.

This stability can be equivalently interpreted as a gradient flow on a convex adaptive landscape. To derive the strictly consistent potential form, we start from the reduced dynamics (20):

$$\frac{dX}{d\tau} = 2X \left[\mathcal{A} \left(1 + \eta \frac{nX^{n-1}}{(1+X^n)^2} \right) - 1 \right] = 2X [\mathcal{A}(1 + \eta h_n(X)) - 1], \quad (41)$$

where $h_n(X) = nX^{n-1}/(1+X^n)^2$ is the cooperative feedback kernel defined earlier. The gradient flow representation $\frac{dX}{d\tau} = -2X \frac{dU}{dX}$ requires the potential derivative to satisfy:

$$\frac{dU}{dX} = 1 - \mathcal{A} - \mathcal{A}\eta h_n(X). \quad (42)$$

Using the exact integral identity $\int h_n(X) dX = \frac{X^n}{1+X^n} + \text{constant}$ (verified by direct differentiation), we obtain the corrected effective potential:

$$U(X) = (1 - \mathcal{A})X - \mathcal{A}\eta \frac{X^n}{1+X^n} + C, \quad (43)$$

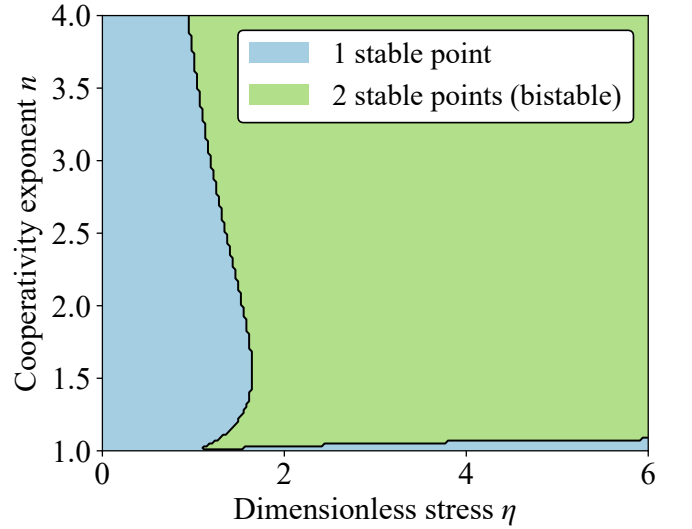


FIG. 4. Phase diagram of the reduced model in the (η, n) parameter space. The color scale indicates the number of steady-state solutions of Eq. (20). A transition from monostability to bistability occurs only for $n > 1$, highlighting the essential role of cooperativity. The bistable region expands with increasing η and n , indicating that stronger stress and higher-order interactions promote multistability and potential switching dynamics.

where C is an arbitrary integration constant. This form strictly satisfies the gradient flow relation, confirming that evolution proceeds downhill on the adaptive landscape with a state-dependent mobility $2X$. The landscape remains strictly convex in the monostable regime, with a single global minimum at X^* , consistent with the Lyapunov stability result.

The curvature of $U(X)$ at the minimum, $\kappa = d^2U/dX^2|_{X=X^*}$, quantifies the strength of restoring selection around the stable fixed point. Using the analytical derivative of the feedback kernel $h'_n(X) = \frac{nX^{n-2}[(n-1)-(n+1)X^n]}{(1+X^n)^3}$, the corrected curvature reads:

$$\kappa = \frac{\mathcal{A}\eta n X^{*n-2} [(n+1)X^{*n} + 1 - n]}{(1+X^{*n})^3}. \quad (44)$$

For $n > 1$ and $X^* > \left(\frac{n-1}{n+1}\right)^{1/n}$ (the stable high-coupling branch), the curvature is positive, consistent with a convex potential minimum. For moderate stress, κ increases with η , reflecting stronger selective pressure for higher coupling under more severe environmental stress. At extremely high stress, $X^* \gg 1$ and $\kappa \sim \eta/X^{*3}$, leading to diminishing selective returns as the protection kernel saturates, consistent with the asymptotic scaling derived in Sec. VI.

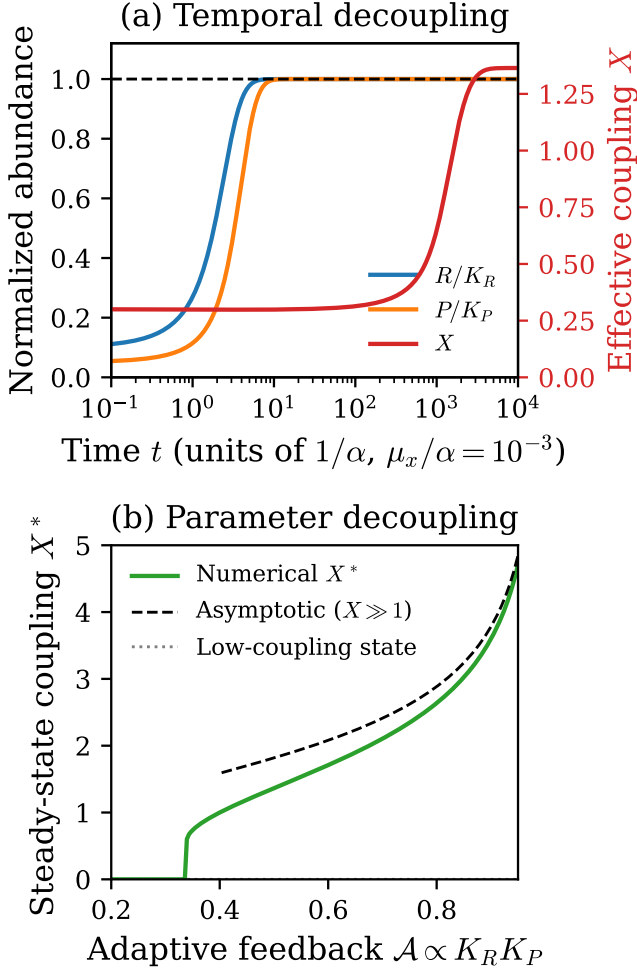


FIG. 5. Functional-abundance decoupling for $n = 2$ cooperative protection and dimensionless stress $\eta = 3$. (a) Temporal decoupling: Normalized RNA (R/K_R , blue) and peptide (P/K_P , orange) abundances saturate to carrying capacities within ~ 10 replication cycles (time t normalized to baseline replication rate α). Effective coupling X (red, right axis), a proxy for collective function, evolves on a $10^3 \times$ slower timescale ($\mu_x/\alpha = 10^{-3}$) with no further change in total abundance after saturation. (b) Parameter decoupling: Steady-state coupling X^* vs dimensionless adaptive feedback $\mathcal{A} \propto K_R K_P$. Below the saddle-node threshold, the system stays in the low-coupling state (gray dotted line); above threshold, X^* jumps to a stable high-coupling branch (green solid line), matching the high- X asymptotic scaling from Eq. (37) (black dashed line). Steady-state function depends only on fixed resource limits, not transient abundance fluctuations.

VIII. CONNECTION TO LIQUID-LIQUID PHASE SEPARATION

We now relate the adaptive dynamics to a coarse-grained thermodynamic description. The mapping from the adaptive variable X to an effective Flory-Huggins interaction parameter is introduced here as a phenomeno-

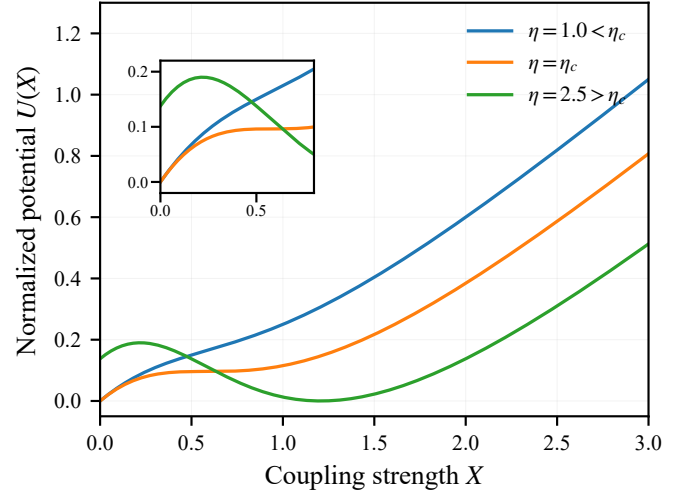


FIG. 6. Corrected adaptive potential $U(X)$ for $n = 2$, $\mathcal{A} = 0.5$, and three values of dimensionless stress η . Potential minima correspond to stable steady states of the reduced dynamics. As stress increases, the high-coupling minimum becomes deeper, consistent with stronger selection for cooperative protection. The inset shows the potential near $X = 0$, where the low-coupling minimum vanishes above the bifurcation threshold η_c .

logical closure for heterotypic contact statistics at the mean-field level; it is not derived microscopically from the underlying coevolutionary kinetics. The purpose of this construction is to examine whether a stress-induced increase in coupling can, within a minimal and bounded closure, shift a free-energy landscape toward the linear instability of the homogeneous state. We do not attempt to describe the dense coacervate phase beyond the spinodal threshold.

Accordingly, the LLPS discussion below should be interpreted as a consistency check on the reduced description and as a coarse-grained instability estimate, not as a direct microscopic derivation of phase separation.

We start from the full ternary Flory-Huggins free energy for an RNA-peptide-solvent mixture, where RNA volume fraction $\phi_R = v_R R$, peptide volume fraction $\phi_P = v_P P$, and solvent volume fraction satisfies $\phi_S = 1 - \phi_R - \phi_P$. Defining polymerization degrees $N_R = v_R/v_0$, $N_P = v_P/v_0$ (where v_0 is a reference monomer volume), the full dimensionless free energy density reads:

$$\frac{f_{\text{full}} v_0}{k_B T} = \frac{\phi_R}{N_R} \ln \phi_R + \frac{\phi_P}{N_P} \ln \phi_P + \phi_S \ln \phi_S - \chi_{RR} \frac{\phi_R^2}{2} - \chi_{PP} \frac{\phi_P^2}{2} - \chi_{\text{eff}} \phi_R \phi_P \quad (45)$$

where χ_{RR} and χ_{PP} are homotypic interaction parameters, and χ_{eff} is the effective heterotypic interaction parameter. We adopt the sign convention where $\chi_{\text{eff}} > 0$ corresponds to net attractive heterotypic interactions, equivalent to $\chi_{\text{FH, standard}} = -\chi_{\text{eff}}$ in the conventional Flory-Huggins sign convention where positive χ indicates unfavorable mixing.

The resource-saturated quasi-steady approximation

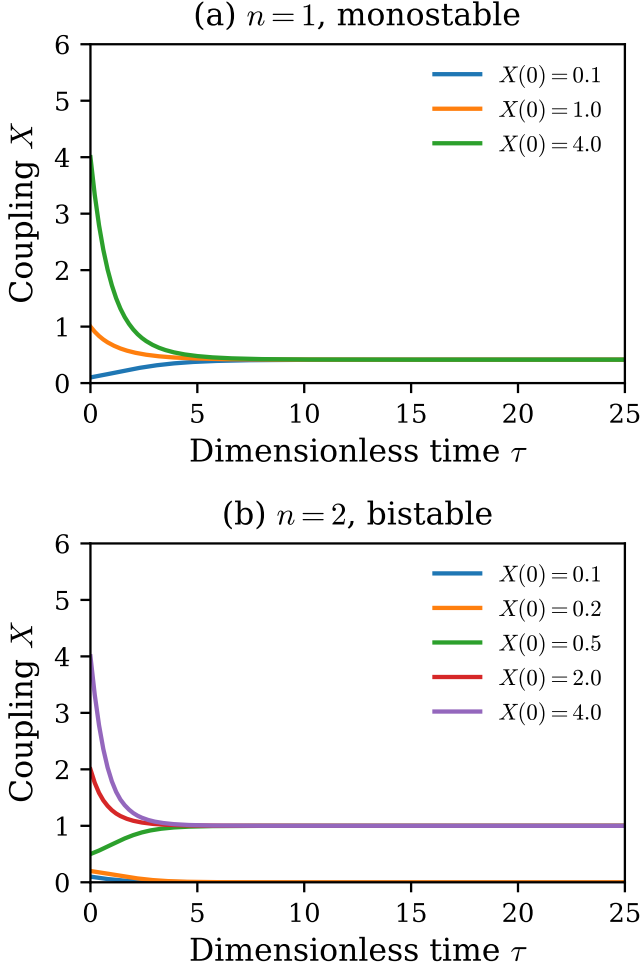


FIG. 7. Time evolution of the effective coupling $X(\tau)$ [Eq. (20)] for $\mathcal{A} = 0.5$ and $\eta = 2$. (a) Noncooperative case ($n = 1$, monostable): all initial conditions $X(0)$ converge to the unique stable positive steady state. (b) Cooperative case ($n = 2$, bistable): initial conditions below the unstable low-coupling branch converge to $X \approx 0$, while those above converge to the stable high-coupling state near $X \approx 1$.

derived in Sec. III ($R \approx K_R$, $P \approx K_P$) enforces a fixed stoichiometric ratio between RNA and peptide volume fractions:

$$s \equiv \frac{\phi_P}{\phi_R} = \frac{v_P K_P}{v_R K_R} = \text{constant}, \quad (46)$$

meaning fluctuations in RNA and peptide concentrations are tightly coupled: $\delta\phi_P = s\delta\phi_R$. Under this fixed-stoichiometry constraint, and for the dilute prebiotic mixture limit $\phi_R + \phi_P \ll 1$ (so $\phi_S \approx 1$, and the solvent entropy term $\phi_S \ln \phi_S$ is approximately constant, with negligible contribution to the stability Hessian), the full free energy reduces to the effective binary form used

in the original draft:

$$\frac{fv_0}{k_B T} \approx \frac{\phi_R}{N_R} \ln \phi_R + \frac{\phi_P}{N_P} \ln \phi_P - \chi_{RR} \frac{\phi_R^2}{2} - \chi_{PP} \frac{\phi_P^2}{2} - \chi_{\text{eff}} \phi_R \phi_P, \quad (47)$$

where all volume fractions are evaluated at their resource-saturated values. The homogeneous state loses stability when the determinant of the reduced Hessian (under fixed stoichiometry) vanishes, yielding the spinodal condition:

$$\left(\frac{1}{N_R \phi_R} - \chi_{RR} \right) \left(\frac{1}{N_P \phi_P} - \chi_{PP} \right) - \chi_{\text{eff}}^2 = 0. \quad (48)$$

This condition is consistent with prior work on associative coacervation of multi-component mixtures with enforced stoichiometric coupling [? ?]. Substituting the fixed resource-saturated volume fractions $\phi_R = v_R K_R$, $\phi_P = v_P K_P$ into Eq. (48) defines a critical heterotypic interaction threshold χ_c required for linear instability of the homogeneous state.

a. Coarse-grained mapping. The monotonic relation between the adaptive coupling variable X and χ_{eff} is introduced as a bounded closure for the renormalization of heterotypic contacts. In the present context, X serves as a proxy for the density of productive RNA-peptide encounters, while χ_{eff} is assumed to increase monotonically with that density. The only structural requirements imposed on the map are monotonicity and saturation: low coupling should correspond to weak effective attraction, while high coupling should approach a finite limiting interaction strength.

To make this assumption explicit, we introduce a generic map

$$\chi_{\text{eff}}(X) = \chi_0 + \Delta\chi \psi(X), \quad (49)$$

with

$$\psi(0) = 0, \quad \psi'(X) > 0, \quad \lim_{X \rightarrow \infty} \psi(X) = 1. \quad (50)$$

These conditions encode only monotonicity and saturation, without committing to a unique microscopic mechanism. The map is therefore best regarded as a coarse-grained parametrization of contact statistics, not as a derived constitutive law.

Within this mean-field closure, the spinodal threshold is crossed whenever

$$0 < \frac{\chi_c - \chi_0}{\Delta\chi} < 1, \quad (51)$$

with the critical coupling X_c defined implicitly by

$$\psi(X_c) = \frac{\chi_c - \chi_0}{\Delta\chi}. \quad (52)$$

A dynamical increase in X that carries the system across X_c signals the onset of linear instability of the homogeneous state. It does not, by itself, determine whether the subsequent evolution proceeds via spinodal decomposition, nucleation, or partial clustering.

b. Representative closure. For explicit calculations, we adopt the saturating form

$$\Theta_X(X) = \frac{X}{X + X_{1/2}}, \quad (53)$$

so that

$$\chi_{\text{eff}}(X) = \chi_0 + \Delta\chi \Theta_X(X), \quad (54)$$

with $X_{1/2}$ a dimensionless half-saturation scale. In the weak-coupling limit,

$$\chi_{\text{eff}}(X) = \chi_0 + \frac{\Delta\chi}{X_{1/2}} X + O(X^2). \quad (55)$$

This choice is a representative member of the admissible class defined in Eq. (49), selected for analytical transparency. The results derived below should therefore be interpreted as properties of this minimal closure, with the qualitative instability criterion expected to persist only within the broad class of bounded monotone maps.

A. Scope of the LLPS closure

B. Scope of the LLPS closure

The mapping from X to χ_{eff} should be interpreted as a phenomenological coarse-graining. Its validity requires that the system remain close to homogeneity and that the trait distributions remain sufficiently narrow for moment closure to apply. In this regime, the spinodal condition identifies only the onset of linear instability of the homogeneous state.

However, crossing the spinodal does not uniquely determine the subsequent morphological pathway. Depending on fluctuations, interfacial tension, finite-size effects, and kinetics, the system may undergo spinodal decomposition, nucleation-and-growth, or only transient clustering. A quantitative description of these processes would require a spatially resolved reaction-diffusion or Cahn-Hilliard-type theory, which lies beyond the scope of the present closure.

Using the representative closure, the critical coupling becomes

$$X_c = X_{1/2} \frac{\chi_c - \chi_0}{\Delta\chi - (\chi_c - \chi_0)}, \quad 0 < \chi_c - \chi_0 < \Delta\chi. \quad (56)$$

Combining this with the high-stress asymptotic scaling

$$X^* \sim (nA\eta)^{1/(n+1)}, \quad (57)$$

yields an estimate of the critical stress required to reach the instability threshold. In the bistable regime ($n > 1$), a saddle-node bifurcation can produce a discontinuous jump in X^* that rapidly moves the system into the linearly unstable region.

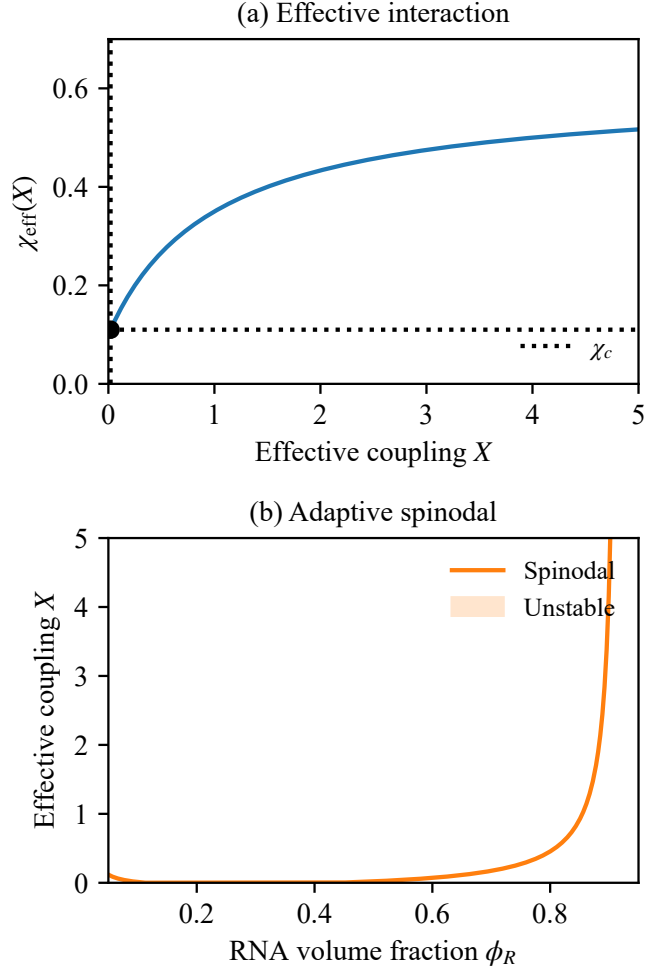


FIG. 8. Coarse-grained link between adaptive coupling X and the linear-instability threshold of a Flory-Huggins description. (a) Effective interaction parameter $\chi_{\text{eff}}(X)$ with representative parameters. The intersection with χ_c defines a critical coupling X_c . (b) Spinodal boundary in the ϕ_R - X plane. The shaded region denotes the linearly unstable region of the homogeneous state; the subsequent demixing pathway depends on spatial and kinetic effects beyond the present model.

a. Interpretation. The present framework does not derive LLPS microscopically. Instead, it provides a minimal coarse-grained mechanism in which a dynamical bifurcation in the adaptive variable can shift the system toward a thermodynamic instability threshold. The subsequent evolution beyond this threshold depends on spatial and non-equilibrium effects that are not included in the reduced description.

IX. HEURISTIC POST-SEPARATION FEEDBACK ENHANCEMENT

Beyond the spinodal threshold, spatial symmetry breaking into a dense coacervate phase (II) and dilute su-

pernatant phase (I) fundamentally alters the evolutionary dynamics. We assume phase equilibrium is established rapidly relative to the slow evolutionary timescale τ , so components partition between phases with fixed equilibrium coefficients. Once the spinodal is crossed, the system demixes into a dilute phase (I) and a dense phase (II). Let f denote the dense-phase volume fraction. Global conservation of material requires

$$R_{\text{tot}} = (1 - f)R^{\text{I}} + fR^{\text{II}}, \quad P_{\text{tot}} = (1 - f)P^{\text{I}} + fP^{\text{II}}. \quad (58)$$

We define partition coefficients

$$\Pi_R \equiv \frac{R^{\text{II}}}{R^{\text{I}}}, \quad \Pi_P \equiv \frac{P^{\text{II}}}{P^{\text{I}}}. \quad (59)$$

The fraction of RNA molecules residing in the dense phase is then

$$\Theta_R = \frac{f\Pi_R}{1 - f + f\Pi_R}. \quad (60)$$

The effective evolutionary force is weighted by Θ_R rather than by volume-weighted averaging, because adaptation is dominated by the actively replicating subpopulation. For strong RNA partitioning ($\Pi_R \gg 1$), $\Theta_R \approx 1$ even for small dense-phase volume fractions $f \ll 1$, meaning nearly all evolutionary change occurs within droplets. If replication is dominated by the dense phase, the local peptide concentration is enhanced by the partition coefficient Π_P , and the adaptive feedback strength $\mathcal{A}$ scales linearly with saturated peptide concentration K_P (per Eq. (19)). The effective feedback parameter is therefore renormalized to

$$\mathcal{A}_{\text{eff}} = \mathcal{A} \frac{\Pi_P}{1 - f + f\Pi_P}, \quad (61)$$

while the local coupling relevant for the protection kernel (which scales with peptide concentration) is amplified to

$$X_{\text{loc}} = X \frac{\Pi_P}{1 - f + f\Pi_P}. \quad (62)$$

The post-separation reduced dynamics therefore takes the form

$$\frac{dX}{d\tau} = 2X \left[\mathcal{A}_{\text{eff}} \left(1 + \eta \frac{nX_{\text{loc}}^{n-1}}{(1 + X_{\text{loc}}^n)^2} \right) - 1 \right]. \quad (63)$$

In the strong-partitioning limit $f\Pi_P \gg 1$, one has $\mathcal{A}_{\text{eff}} \approx \mathcal{A}/f$, suggesting that compartmentalization can enhance the local feedback strength experienced by the protected RNA-peptide complexes. However, Eqs. (58)–(63) should be interpreted only as a heuristic coarse-grained extension of the pre-separation theory. Once demixing has occurred, interfacial transport, phase-specific kinetics, and spatial gradients generally prevent the single-variable description in terms of X from remaining strictly closed. The post-separation argument therefore supports

only the qualitative statement that partitioning may reinforce local adaptive feedback; a quantitatively predictive theory would require an explicitly spatial reaction–diffusion or Cahn–Hilliard-type treatment.

Despite these limitations, the coarse-grained model predicts a robust scaling behavior for post-separation adaptation: in the strong-partitioning limit $f\Pi_P \gg 1$, the effective feedback strength scales as $\mathcal{A}_{\text{eff}} \sim \mathcal{A}/f$, leading to an evolutionary velocity $dX/d\tau \sim 1/f$. This implies that smaller, more dilute droplets (lower f) accelerate adaptation more strongly, consistent with prebiotic scenarios where wet-dry cycles generate transient small-volume condensates. The resulting accelerated feedback can drive the rapid evolution of higher coupling strengths, potentially enabling the emergence of primitive, translation-like peptide synthesis activity within condensed proto-ribosomal assemblies.

X. SPATIAL EFFECTS AND LIMITS OF THE WELL-MIXED DESCRIPTION

The reduced model is derived under a homogeneous approximation. To delineate its validity, it is useful to compare reaction and diffusion timescales over an observation length L . Let D_R and D_P denote effective diffusivities. The relevant diffusion times are

$$\tau_{\text{diff}}^{(R)} \sim \frac{L^2}{D_R}, \quad \tau_{\text{diff}}^{(P)} \sim \frac{L^2}{D_P}. \quad (64)$$

The well-mixed reduction is controlled when

$$\frac{\alpha L^2}{D_R} \ll 1, \quad \frac{\mu_y L^2}{D_P} \ll 1, \quad (65)$$

so that local gradients relax faster than reaction and trait-adaptation processes. Once these conditions fail, or once LLPS has already set in, X should be interpreted only as a coarse-grained indicator rather than a fully local state variable.

XI. STOCHASTIC MONTE CARLO SIMULATION OF SPATIAL RNA-PEPTIDE EVOLUTION

To benchmark the reduced mean-field description against a stochastic particle model, we perform a two-dimensional Monte Carlo simulation of an RNA-peptide mixture in a square domain with reflecting boundaries. The simulation is intentionally minimal but fully specified: every realization evolves under the same update protocol, and all random events are generated from a seeded pseudorandom number generator. The goal is not to reproduce a particular experimental system quantitatively, but to test whether the reduced model captures the dominant coarse-grained trend emerging from stochastic microscopic dynamics.

TABLE I. Monte Carlo simulation protocol used for the stochastic RNA–peptide benchmark. The values shown here are those used in the code for the main validation run.

Quantity	Value / rule
Domain size	$L = 10$
Initial populations	$N_{\text{RNA}}(0) = 40, N_{\text{PEP}}(0) = 40$
Boundary condition	Reflecting
Total steps	1800
Random seed	Fixed per realization
Diffusion	Trait-independent Gaussian displacement
Bound diffusion factor	0.35
Binding radius	$r_b = 0.7$
Unbinding rule	Stress-enhanced stochastic dissociation
Crowding control	Logistic factor $[1 + \exp(\dots)]^{-1}$
Cluster radius	0.85
Snapshot times	0, 200, 500, 800, 1000

The system contains two particle species, RNA and peptide, each carrying a continuous trait variable in $[0, 1]$. The initial population is drawn uniformly in space, with traits sampled from narrow bounded distributions. The particles move in a finite box of side length $L = 10$, and spatial confinement is implemented by reflecting boundary conditions. Bound states between RNA and peptide particles are allowed, and bound complexes experience reduced diffusion together with a weak local tightening force that keeps complexes spatially compact.

At each Monte Carlo step, the dynamics are updated in the following order: (i) diffusion, (ii) unbinding, (iii) binding, (iv) RNA turnover, (v) peptide turnover, and (vi) local tightening of bound complexes. The update rules are summarized in Table I. Binding is restricted to nearby RNA–peptide pairs within a finite interaction radius, and the binding probability depends on the particle traits, the environmental stress, and a smooth crowding factor. Unbinding is treated stochastically with a stress-enhanced dissociation probability that decreases with pair stability. Birth and death events are then applied separately to RNA and peptide species, with protection and helper effects mediated by the current bound partner. Trait inheritance includes small Gaussian mutation and a weak drift toward the partner trait, thereby allowing slow coevolution of both sequence-like and functional variables.

To characterize the simulation output, we record the time series of RNA abundance, peptide abundance, number of bound pairs, and mean trait values. In addition, we sample cluster statistics at regular intervals by constructing connected components from both spatial proximity and binding links. This provides a coarse-grained measure of aggregation beyond pairwise binding alone.

For completeness, the Monte Carlo implementation uses the following acceptance rules. If r and p denote the RNA and peptide traits of a candidate pair, the binding

probability is of the form

$$p_{\text{bind}} = \min \left\{ 0.95, p_0^{(b)} (0.2 + r)(0.2 + p) [1 + \sigma_b s] (0.7 + 0.3 \phi_{\text{crowd}}) \right\} \quad (66)$$

where s is the environmental stress and ϕ_{crowd} is the crowding factor. The unbinding probability is

$$p_{\text{unbind}} = p_0^{(u)} \frac{1 + s}{\max[0.25, 1 + 2\bar{t}]}, \quad (67)$$

where $\bar{t}$ is the mean trait of the bound pair. Birth and death rates are modulated by protection and helper terms of the form

$$p_{\text{birth}} \propto (\text{baseline}) \times (\text{trait factor}) \times (\text{partner assistance}) \times (\text{crowding}) \quad (68)$$

$$p_{\text{death}} \propto (\text{baseline}) \times (\text{stress factor}) \times (\text{lack of protection}) \times (\text{trait comp}) \quad (69)$$

These rules define the stochastic microscopic benchmark against which the reduced model is fitted.

The simulation outputs are saved as time-series plots, spatial snapshots, density heat maps, and cluster-size distributions. The main purpose of these observables is to provide a reproducible comparison with the reduced dynamics, rather than to represent a complete mechanistic description of any specific biological system.

XII. DISCUSSION

The central physical result of this work is that cooperative protection under stress can induce a saddle-node bifurcation in the effective coupling of an RNA–peptide mixture. When the Hill exponent satisfies $n = 1$ (noncooperative case), the system exhibits a smooth, monostable response to increasing stress. For $n > 1$, the nonlinear feedback kernel develops an interior maximum, leading to a saddle-node bifurcation at a critical stress $\eta_c = (1/\mathcal{A} - 1)/C_n$. Above this threshold, the system undergoes a discontinuous transition to a high-coupling state whose steady-state value scales as $X^* \sim (n\mathcal{A}\eta)^{1/(n+1)}$ in the strong-stress limit.

A key consequence of the underlying timescale separation is the complete decoupling between functional evolution and molecular abundance. Total RNA and peptide concentrations rapidly saturate to their carrying capacities on fast replication timescales, while the coupling coordinate X —which encodes the collective functional quality—evolves on a much slower mutational timescale. This functional-abundance decoupling is a generic feature of resource-limited adaptive systems and implies that functional improvement does not require continuous population growth.

By mapping the dynamical variable X to an effective heterotypic interaction parameter $\chi_{\text{eff}}(X)$ through a bounded monotone closure, we show that the stress-induced jump in coupling can bring the mixture close to the spinodal boundary of a Flory–Huggins description.

This suggests a plausible two-step scenario in which dynamical adaptation precedes and potentially facilitates liquid–liquid phase separation. Stochastic Monte Carlo simulations of a spatially resolved particle-based model confirm that the reduced one-dimensional description captures the dominant coarse-grained dynamics, including the stress-driven enhancement of heterotypic contacts.

While the precise location of the bifurcation threshold and the spinodal condition depend on the details of the chosen closure, the qualitative mechanism—cooperativity converting a continuously increasing environmental stress into a switch-like transition in both function and organization—is robust. The present framework thus offers a physically transparent minimal model for understanding how nonequilibrium driving, cooperative interactions, and thermodynamic instability may interact in prebiotic RNA–peptide systems. Future extensions incorporating explicit spatial dynamics or higher-moment corrections near the bifurcation will help quantify the robustness of this scenario beyond the current mean-field reduction.

Appendix A: Notation and dimensional consistency

Table II collects the symbols used in the manuscript, together with their physical roles, dimensions, and the stage at which they first enter the theory. This format is intended to make the reduction chain transparent and to avoid repeated redefinitions in the main text.

The reduction is organized so that each dimensionless control parameter appears only once in the main derivation. In particular, X , τ , $\mathcal{A}$, and η are introduced in Sec. III and then used consistently thereafter without redefinition. The thermodynamic closure in Sec. V introduces $\chi_{\text{eff}}(X)$ only once, as a phenomenological mapping from the adaptive coordinate to the coarse-grained interaction parameter.

Appendix B: Moment Closure and Mean-Trait Dynamics

To derive the mean-trait dynamics used in the main text, we work at the level of the first moments of the trait distributions. Let $R(x, t)$ denote the RNA concentration density over the dimensionless trait x , and define

$$R(t) = \int R(x, t) dx, \quad \bar{x}(t) = \frac{1}{R(t)} \int x R(x, t) dx. \quad (\text{B1})$$

The RNA trait-structured dynamics is taken to be

$$\partial_t R(x, t) = [r(x, t) - d(x, t)]R(x, t) + D_x \partial_x^2 R - \mu_x (x - x_0)R, \quad (\text{B2})$$

TABLE II. Symbols, meanings, and dimensions used in the manuscript. Here C denotes concentration, T denotes time, and 1 denotes a strictly dimensionless quantity. The column “First introduced” indicates where the symbol first appears in the derivation.

Symbol	Meaning	Dimension	First introduced	intro-
R	Total RNA concentration	C	Sec. II	
P	Total peptide concentration	C	Sec. II	
K_R	RNA carrying capacity	C	Sec. II	
K_P	Peptide carrying capacity	C	Sec. II	
t	Physical time	T	Sec. II	
$\bar{x}$	Mean RNA binding trait	1	Sec. II	
$\bar{y}$	Mean peptide protective trait	1	Sec. II	
α	RNA baseline replication rate	T^{-1}	Sec. II	
δ_0	Baseline degradation rate	T^{-1}	Sec. II	
k_0	Basal peptide production rate	T^{-1}	Sec. II	
μ	Peptide turnover/clearance rate	T^{-1}	Sec. II	
μ_x	RNA trait relaxation rate	T^{-1}	Sec. II	
μ_y	Peptide trait relaxation rate	T^{-1}	Sec. II	
β	Replication coupling coefficient	C^{-1}	Sec. II	
γ	Protection coupling coefficient	C^{-1}	Sec. II	
κ	Trait-dependent peptide production coupling	$C^{-1}T^{-1}$	Sec. II	
E	Environmental stress load	1	Sec. II	
n	Hill cooperativity exponent	1	Sec. III	
V_x	RNA trait variance	1	Sec. III	
λ	Peptide adiabatic slaving coefficient	1	Sec. III	
X	Reduced effective coupling variable	1	Sec. III	
τ	Reduced slow time	1	Sec. III	
$\mathcal{A}$	Reduced baseline adaptive feedback	1	Sec. III	
η	Reduced stress load parameter	1	Sec. III	
χ_0	Baseline Flory–Huggins interaction	1	Sec. V	
$\Delta\chi$	Maximum interaction parameter shift	1	Sec. V	
χ_{eff}	Effective interaction parameter	1	Sec. V	
χ_c	Critical interaction parameter (spinodal)	1	Sec. V	
$X_{1/2}$	Half-saturation scale for coupling	1	Sec. V	

with

$$r(x, t) = \alpha[1 + \beta x P(t) \bar{y}(t)], \quad d(x, t) = \frac{\delta_0 E}{1 + [\gamma x P(t) \bar{y}(t)]^n}. \quad (\text{B3})$$

Standard moment manipulations give

$$\frac{dR}{dt} = \langle r - d \rangle_R R, \quad (\text{B4})$$

and

$$\frac{d\bar{x}}{dt} = \text{Cov}_R(x, r - d) - \mu_x(\bar{x} - x_0). \quad (\text{B5})$$

Under the narrow-distribution approximation, the covariance term closes as

$$\text{Cov}_R(x, r - d) \approx V_x \left. \frac{\partial(r - d)}{\partial x} \right|_{x=\bar{x}}, \quad (\text{B6})$$

which yields, after setting $x_0 = 0$,

$$\frac{d\bar{x}}{dt} = V_x \left[\alpha\beta P\bar{y} + \frac{\delta_0 E n \gamma^n P^n \bar{y}^n \bar{x}^{n-1}}{[1 + (\gamma\bar{x}P\bar{y})^n]^2} \right] - \mu_x \bar{x}. \quad (\text{B7})$$

For the peptide sector, the reduced theory does not require a second full trait-resolved partial differential equation. It is sufficient to assume that the first moment obeys the minimal linear closure

$$\frac{d\bar{y}}{dt} = \kappa V_y R\bar{x} - \mu_y \bar{y}, \quad (\text{B8})$$

which is the lowest-order form consistent with (i) promotion of peptide quality by coupling to the RNA sector and (ii) relaxation of the peptide trait on a faster timescale. In the quasi-steady regime $\mu_y \gg \mu_x$ and $R \simeq K_R$, Eq. (B8) gives

$$\bar{y} \simeq \lambda \bar{x}, \quad \lambda = \frac{\kappa V_y K_R}{\mu_y}. \quad (\text{B9})$$

Substituting this relation into Eq. (B7) produces the scalar reduction summarized in Appendix D.

1. Full Second-Order Moment Closure

To quantify deviations from the narrow-trait-distribution assumption, we extend the moment-closure framework to include the evolution of the RNA trait variance V_x . Starting from the trait-resolved dynamics for $R(x, t)$ given in Eq. (B1), the variance satisfies the standard evolution equation under the diffusion approximation:

$$\frac{dV_x}{dt} = 2\text{Cov}_R(x, r - d) + D_x - 2\mu_x V_x, \quad (\text{B10})$$

where D_x is the dimensionless mutational diffusivity, and the covariance term is again closed at leading order as

$$\text{Cov}_R(x, r - d) \approx V_x \left. \frac{\partial(r - d)}{\partial x} \right|_{x=\bar{x}}. \quad (\text{B11})$$

Substituting the expressions for r and d from Eq. (B2), we obtain

$$\frac{dV_x}{dt} = 2V_x \left[\alpha\beta P\bar{y} + \frac{\delta_0 E n \gamma^n P^n \bar{y}^n \bar{x}^{n-1}}{[1 + (\gamma\bar{x}P\bar{y})^n]^2} \right] + D_x - 2\mu_x V_x. \quad (\text{B12})$$

The full second-order moment-closure model consists of Eqs. (B3), (B4), and (B12), solved simultaneously under the same quasi-steady assumptions $R \simeq K_R$, $P \simeq K_P$ that underpin the scalar reduction. This model allows the trait variance to evolve in response to selection gradients, eliminating the fixed- V_x assumption of the reduced one-dimensional description.

Appendix C: Timescale Separation and One-Dimensional Reduction

The full trait-structured model exhibits a clear hierarchy of timescales. In typical prebiotic contexts, replication and degradation are fast ($\alpha, \delta_0 \sim \mathcal{O}(1 \text{ h}^{-1})$), peptide turnover is intermediate ($\mu_y \sim \mathcal{O}(10^{-1} \text{ h}^{-1})$), and RNA trait evolution is slow ($\mu_x \sim \mathcal{O}(10^{-4} \text{ h}^{-1})$). This separation, $\mu_x \ll \mu_y \ll \alpha$, justifies two adiabatic reductions:

a. 1. Population Saturation. Because population growth outpaces trait evolution, total concentrations rapidly approach their carrying capacities enforced by logistic regulation. We thus set $R \simeq K_R$ and $P \simeq K_P$.

b. 2. Adiabatic Slaving of Peptide Trait. Using the mean-trait closure in Eq. (B8) and imposing quasi-steady relaxation on the fast timescale, $d\bar{y}/dt \approx 0$, we obtain

$$\bar{y} \simeq \lambda \bar{x}, \quad \lambda = \frac{\kappa V_y K_R}{\mu_y}. \quad (\text{C1})$$

c. 3. Reduction to Dimensionless Coupling X . Substituting $P = K_P$ and $\bar{y} = \lambda \bar{x}$ into Eq. (B7), the RNA trait dynamics becomes

$$\frac{d\bar{x}}{dt} = V_x \left[\alpha\beta K_P \lambda \bar{x} + \frac{\delta_0 E n \gamma^n K_P^n \lambda^n \bar{x}^{2n-1}}{[1 + (\gamma K_P \lambda \bar{x}^2)^n]^2} \right] - \mu_x \bar{x}. \quad (\text{C2})$$

We introduce the dimensionless state variable

$$X \equiv \gamma K_P \lambda \bar{x}^2 \quad (\text{C3})$$

and the dimensionless slow time

$$\tau \equiv \mu_x t. \quad (\text{C4})$$

By the chain rule,

$$\frac{dX}{d\tau} = \frac{1}{\mu_x} \frac{dX}{dt} = \frac{2\gamma K_P \lambda \bar{x}}{\mu_x} \frac{d\bar{x}}{dt}. \quad (\text{C5})$$

Multiplying Eq. (C2) by $\frac{2\gamma K_P \lambda \bar{x}}{\mu_x}$ yields

$$\frac{dX}{d\tau} = 2X \left[\frac{\alpha\beta V_x K_P \lambda}{\mu_x} + \frac{\delta_0 E V_x K_P \lambda}{\mu_x} \frac{n(\gamma K_P \lambda \bar{x}^2)^{n-1}}{[1 + (\gamma K_P \lambda \bar{x}^2)^n]^2} - 1 \right]. \quad (\text{C6})$$

Recognizing X in the nonlinear term and defining the strictly dimensionless control parameters

$$\mathcal{A} \equiv \frac{\alpha\beta V_x K_P \lambda}{\mu_x}, \quad \eta \equiv \frac{\gamma\delta_0 E}{\alpha\beta}, \quad (\text{C7})$$

we observe that the prefactor of the second term is exactly $\mathcal{A}\eta$. This exactly recovers the reduced scalar equation presented in the main text:

$$\frac{dX}{d\tau} = 2X \left[\mathcal{A} \left(1 + \eta \frac{nX^{n-1}}{(1+X^n)^2} \right) - 1 \right]. \quad (\text{C8})$$

Appendix D: Large- X Asymptotics

When $X \gg 1$, Eq. (20) reduces at steady state to

$$\mathcal{A}\eta \frac{n}{X^{n+1}} \simeq 1, \quad (\text{D1})$$

which directly yields the scaling relation in Eq. (37).

Appendix E: Validation of the reduced model

To assess the accuracy of the one-dimensional reduced description derived in Sec. III, we directly compare its dynamics with those of the full moment-closure system introduced in Appendix C.

We consider the reduced variable

$$X = \gamma K_P \lambda \bar{x}^2, \quad (\text{E1})$$

and integrate both the full system and the reduced equation

$$\frac{dX}{d\tau} = 2X \left[\mathcal{A} \left(1 + \eta \frac{nX^{n-1}}{(1+X^n)^2} \right) - 1 \right], \quad (\text{E2})$$

where $\tau = \mu_x t$, using identical parameter values and consistent initial conditions.

Figure 10 shows the resulting trajectories $X(\tau)$ obtained from the two descriptions. The reduced model quantitatively reproduces the full dynamics over the entire relaxation process, including both the transient approach and the steady-state value.

This agreement confirms that the timescale separation and moment-closure approximation leading to the reduced equation provide an accurate description of the slow evolutionary dynamics. In particular, deviations remain negligible throughout the trajectory, indicating that higher-order corrections do not qualitatively affect the behavior in the parameter regime considered.

Appendix F: Minimal Cahn–Hilliard Extension for LLPS

To complement the coarse-grained spinodal criterion used in the main text, we now introduce a minimal spatially extended description of liquid–liquid phase separation (LLPS) in the form of a Cahn–Hilliard equation.

The purpose of this appendix is not to provide a quantitatively complete theory of coacervation, but to illustrate how the effective interaction parameter $\chi_{\text{eff}}(X)$ may be embedded into a standard conserved-order-parameter dynamics within the same phenomenological closure.

We consider a conserved composition field $\phi(\mathbf{r}, t)$ and define the coarse-grained free-energy functional

$$\mathcal{F}[\phi] = \int d\mathbf{r} \left[f(\phi; X) + \frac{\kappa_\phi}{2} |\nabla \phi|^2 \right], \quad (\text{F1})$$

where $\kappa_\phi > 0$ is a gradient-energy coefficient and

$$f(\phi; X) = \frac{\phi}{N_R} \ln \phi + \frac{1-\phi}{N_P} \ln(1-\phi) + \chi_{\text{eff}}(X) \phi(1-\phi). \quad (\text{F2})$$

Here N_R and N_P are the effective polymerization indices of the two components, and $\chi_{\text{eff}}(X)$ is the same bounded monotone closure introduced in the main text.

The conserved dynamics is then given by

$$\frac{\partial \phi}{\partial t} = \nabla \cdot (M \nabla \mu), \quad \mu(\mathbf{r}, t) = \frac{\delta \mathcal{F}}{\delta \phi}, \quad (\text{F3})$$

with constant mobility $M > 0$. Explicitly,

$$\mu = \frac{1}{N_R} (\ln \phi + 1) - \frac{1}{N_P} (\ln(1-\phi) + 1) + \chi_{\text{eff}}(X) (1-2\phi) - \kappa_\phi \nabla^2 \phi. \quad (\text{F4})$$

Linearizing around a spatially homogeneous state $\phi(\mathbf{r}, t) = \phi_0 + \delta\phi(\mathbf{r}, t)$ with small perturbations $\delta\phi$, one obtains for a Fourier mode $\delta\phi_{\mathbf{k}}$ the growth rate

$$\omega(k) = -Mk^2 [f''(\phi_0; X) + \kappa_\phi k^2], \quad (\text{F5})$$

where

$$f''(\phi_0; X) = \frac{1}{N_R \phi_0} + \frac{1}{N_P (1-\phi_0)} - 2\chi_{\text{eff}}(X). \quad (\text{F6})$$

Therefore, the homogeneous state is linearly unstable when

$$f''(\phi_0; X) < 0, \quad (\text{F7})$$

which is precisely the spinodal criterion used in the main text. The corresponding fastest-growing wavelength follows from Eq. (F5) as

$$k_*^2 = -\frac{f''(\phi_0; X)}{2\kappa_\phi}, \quad \lambda_* = \frac{2\pi}{k_*}, \quad (\text{F8})$$

provided that $f''(\phi_0; X) < 0$.

This minimal Cahn–Hilliard extension shows that the effective coupling variable X does not merely shift an equilibrium threshold, but can also control the spatial scale and growth rate of the initial demixing instability. At the same time, it remains a coarse-grained description: kinetic arrest, nonlinear droplet coarsening, hydrodynamic effects, and explicit nonequilibrium driving are not included here and would require a more complete theory.

Appendix G: Optimized Fitting Pipeline for Monte Carlo Validation

To compare the stochastic simulation with the reduced equation, we construct an effective scalar observable from each Monte Carlo trajectory and fit the reduced model to the ensemble-averaged dynamics. The comparison is designed to be conservative: the fitting protocol is fixed in advance, the preprocessing steps are explicit, and the role of each additional numerical parameter is stated clearly.

For each Monte Carlo realization, we discard an initial transient of 50 steps and then compute the effective coupling observable

$$X_{\text{MC}}(t) = \frac{N_{\text{pair}}(t)}{\min[N_{\text{RNA}}(t), N_{\text{PEP}}(t)] + \epsilon} \bar{x}(t) \bar{y}(t), \quad (\text{G1})$$

where N_{pair} is the number of bound pairs, $\bar{x}$ is the mean RNA trait, $\bar{y}$ is the mean peptide trait, and ϵ is a small regularization constant used only to prevent division by zero. This scalar proxy is intended to capture the combined effect of binding abundance and trait-level cooperation in a single coarse-grained coordinate.

We generate $n_{\text{rep}} = 5$ independent Monte Carlo realizations with different random seeds and align them on a common discrete time grid. The trajectories are averaged pointwise across repeats and then smoothed with a three-point moving average. The smoothing is not a model assumption; it is used only to reduce short-time stochastic noise in the ensemble mean. The fitting is performed on the smoothed ensemble average, not on individual noisy trajectories.

The reduced model is the closed scalar equation

$$\frac{dX}{d\tau} = 2X \left[\mathcal{A} \left(1 + \eta \frac{nX^{n-1}}{(1+X^n)^2} \right) - 1 \right], \quad (\text{G2})$$

where $\mathcal{A}$ is the baseline growth parameter, η is the stress-controlled cooperative enhancement, and n is the apparent Hill exponent. In the fitting pipeline, an additional scale factor α_{scale} is introduced to convert the normalized Monte Carlo time variable into the dimensionless reduced-model time. This factor is treated as a nuisance calibration parameter: it aligns the time axis but does not introduce a new physical mechanism. The fitted curve therefore probes the shape of the reduced dynamics, while α_{scale} absorbs the unknown mapping between Monte Carlo steps and coarse-grained time units.

The fitting objective is a weighted least-squares residual,

$$\mathcal{L} = \sum_i w_i [X_{\text{red}}(t_i; \mathcal{A}, \eta, n, \alpha_{\text{scale}}) - X_{\text{MC}}(t_i)]^2, \quad (\text{G3})$$

with piecewise constant weights

$$w_i = \begin{cases} 2, & \tau_i/\tau_{\text{max}} \leq 0.3, \\ 1, & \tau_i/\tau_{\text{max}} > 0.3. \end{cases} \quad (\text{G4})$$

TABLE III. Fitting pipeline used for Monte Carlo validation. The values listed here correspond to the main benchmark reported in the manuscript.

Quantity	Value / rule
Independent repeats	$n_{\text{rep}} = 5$
Burn-in discarded	50 steps
Smoothing window	3-point moving average
Weighted region	$\tau/\tau_{\text{max}} \leq 0.3$
Early-time weight	$w_i = 2$
Late-time weight	$w_i = 1$
Fit parameters	$\mathcal{A}, \eta, n, \alpha_{\text{scale}}$
Objective function	Weighted least squares
Error metrics	Mean relative error, weighted R^2

The early-time upweighting is a pragmatic choice motivated by the fact that the transient regime contains the strongest signal for distinguishing growth laws; it should be interpreted as a numerical robustness device rather than as a physical reweighting principle.

The optimization is carried out with constrained nonlinear least squares, and the fitted parameters are extracted from the best-fit solution. To quantify the agreement, we report both the mean relative error

$$\delta = \frac{1}{N} \sum_i \frac{|X_{\text{MC}}(t_i) - X_{\text{red}}(t_i)|}{1 + \max[X_{\text{MC}}(t_i), X_{\text{red}}(t_i)]}, \quad (\text{G5})$$

and the weighted coefficient of determination

$$R_w^2 = 1 - \frac{\sum_i w_i [X_{\text{MC}}(t_i) - X_{\text{red}}(t_i)]^2}{\sum_i w_i [X_{\text{MC}}(t_i) - \bar{X}_w]^2}, \quad (\text{G6})$$

where $\bar{X}_w$ is the weighted mean of the Monte Carlo data.

Table III summarizes the numerical settings used in the validation run.

For the benchmark run, the optimization returns a low effective growth coefficient, a strong cooperative enhancement, and a saturated apparent Hill exponent at its lower bound when the noncooperative limit is not enforced. The resulting fit reproduces the ensemble-averaged Monte Carlo trajectory with sub-percent mean relative error and a high weighted R^2 , indicating that the reduced equation captures the dominant coarse-grained trend under the adopted closure. At the same time, the auxiliary scale factor α_{scale} and the weighting prescription should be regarded as part of the validation protocol, not as additional mechanistic predictions.

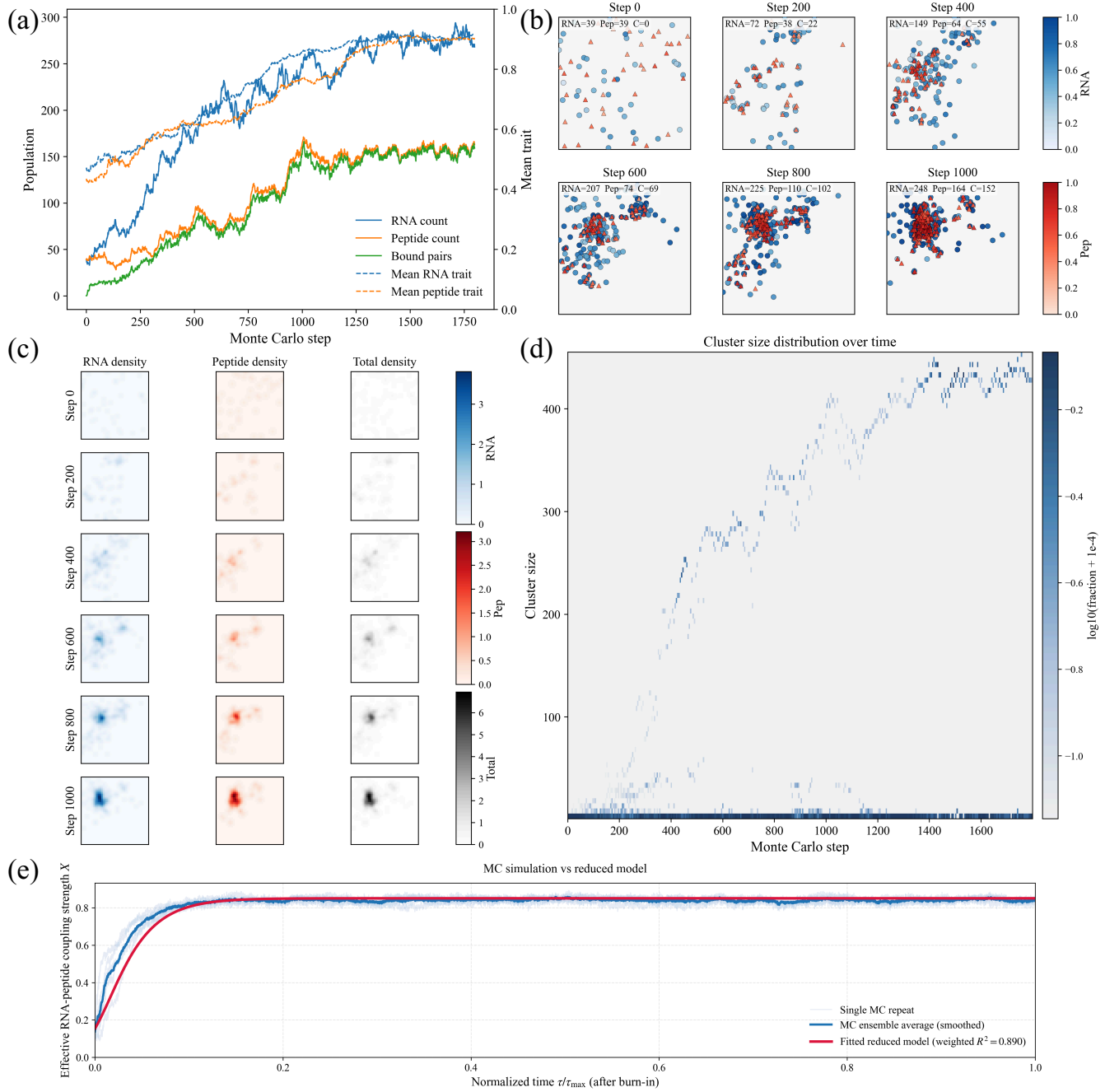


FIG. 9. (a) Temporal evolution of the stochastic RNA-peptide Monte Carlo simulation. The total numbers of RNA molecules, peptides, and bound RNA-peptide pairs are shown together with the mean RNA catalytic trait and the mean peptide binding/protection trait. The simulation illustrates how stochastic diffusion, binding/unbinding, replication, mutation, and stress-dependent turnover reshape both abundance and functional quality over time. (b) Representative spatial slices of the Monte Carlo dynamics at selected time points. RNA molecules are shown as circles and peptides as triangles, with color gradients indicating catalytic trait values for RNA and binding/protection trait values for peptides. (c) Spatial density heatmaps for RNA, peptides, and the combined molecular population at selected simulation times. The maps quantify spatial heterogeneity and reveal whether molecular redistribution remains diffuse or develops concentrated high-density regions. These heatmaps provide a compact visualization of mesoscale organization beyond point-like particle plots. (d) Time-resolved cluster statistics extracted from the Monte Carlo simulation. This panel shows the cluster-size distribution as a function of time, where cluster sizes are defined by connected components formed through binding links and short-range spatial proximity. (e) Evolution of the effective coupling X from spatial Monte Carlo simulations (blue) and the fitted reduced model (red). Thin lines show individual stochastic realizations, and the thick blue curve denotes the ensemble average after burn-in removal and smoothing. The reduced model closely reproduces the averaged dynamics, indicating that it captures the dominant feedback behavior of the stochastic system.

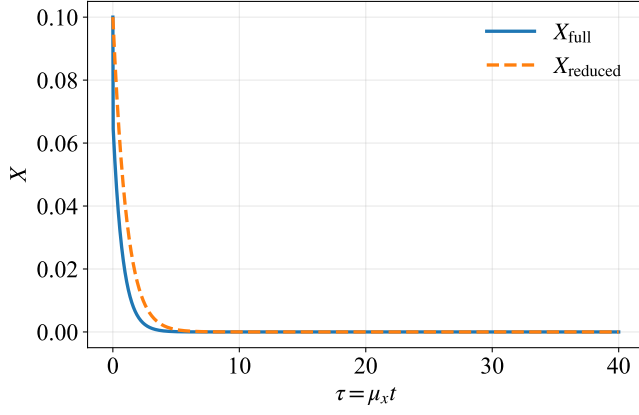


FIG. 10. Comparison between the full moment-closure model and the reduced one-dimensional description. The solid line shows the evolution of $X(\tau)$ computed from the full system, while the dashed line corresponds to the reduced equation. Both models are integrated using identical parameters and consistent initial conditions. The close agreement over the entire trajectory demonstrates the quantitative accuracy of the reduced model.

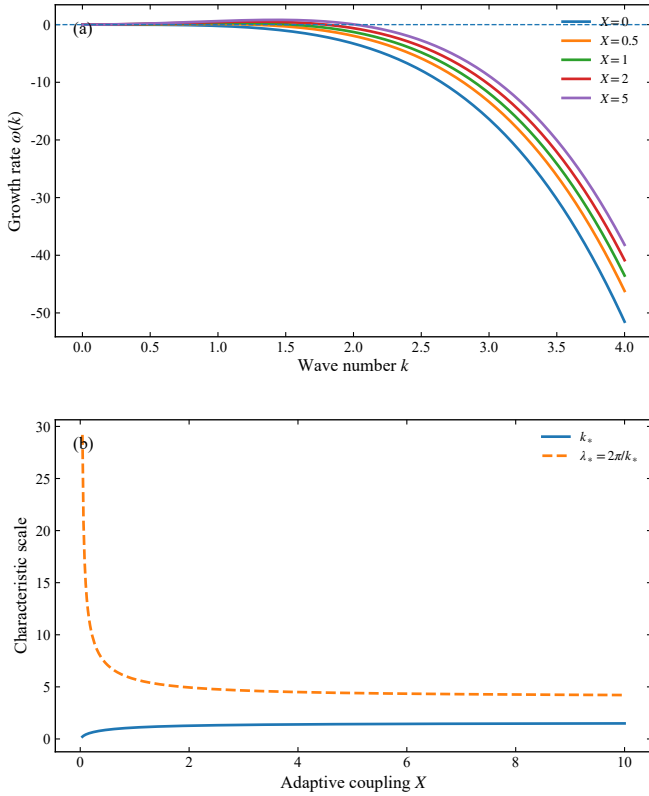


FIG. 11. Minimal Cahn–Hilliard extension of the coarse-grained LLPS mapping. (a) Dispersion relation $\omega(k)$ for several values of the adaptive coupling X . When $f''(\phi_0; X) < 0$, long-wavelength modes become unstable, indicating linear instability of the homogeneous state. (b) Characteristic instability scale extracted from the fastest-growing mode k_* and the corresponding wavelength $\lambda_* = 2\pi/k_*$ as functions of X .